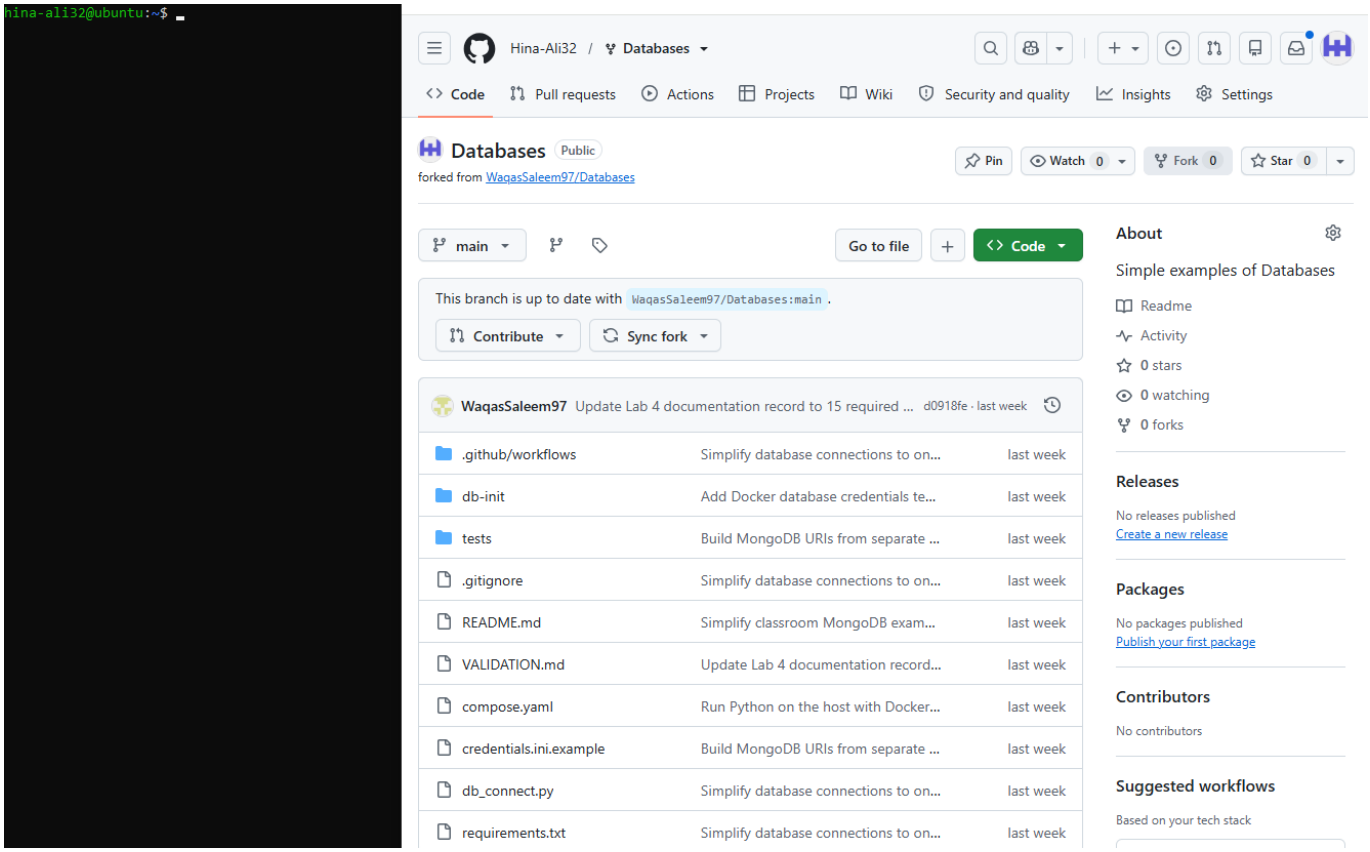
Fatima Jinnah Women University



Department of Computer Science
DevOps Lab # 4

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Registration no: 2023-BCS-040
Course code: CS-649
Section: BCS VII(B)
GitHub Username: Hina-Ali32

Task 1: Fork and Clone the Databases Repository



Shows the fork under the personal GitHub account next to the original Databases repository, confirming the fork source.

```
hina-ali32@ubuntu:~$ cd ~
hina-ali32@ubuntu:~$ git clone git@github.com:Hina-Ali32/Databases.git
Cloning into 'Databases'...
remote: Enumerating objects: 101, done.
remote: Counting objects: 100% (101/101), done.
remote: Compressing objects: 100% (90/90), done.
remote: Total 101 (delta 30), reused 1 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (101/101), 39.86 KiB | 887.00 KiB/s, done.
Resolving deltas: 100% (30/30), done.
hina-ali32@ubuntu:~$ cd Databases
hina-ali32@ubuntu:~/Databases$
```

Shows the fork cloned over SSH into ~/Databases on the Ubuntu server.

Task 2: Install Docker, Docker Compose, and Python on Ubuntu

```
hina-ali32@ubuntu:~$ git --version
git version 2.53.0
hina-ali32@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
hina-ali32@ubuntu:~$ docker compose version
Docker Compose version v5.6.1
hina-ali32@ubuntu:~$
```

Shows Git, Docker, and Docker Compose installed and reporting their versions after reconnecting to the Ubuntu server.

```
hina-ali32@ubuntu:~$ sudo apt install -y python3 python3-venv python3-pip
[sudo: authenticate] Password:
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Installing:
  python3-pip  python3-venv

Installing dependencies:
  build-essential  cpp-15-x86-64-linux-gnu  g++                    gcc                    gcc-x86-64-linux-gnu      libasan8      libfakeroot            libhwasan0  libmpc3          libubsan1          python3-setuptools-whl
  bzip2            cpp-x86-64-linux-gnu     g++-15                 gcc-15                 libalgorithm-diff-perl    libcc1-0      libfile-fcntllock-perl  libisl23    libquadmath0     lto-disabled-list  python3-wheel
  cpp              dpkg-dev                 g++-15-x86-64-linux-gnu  gcc-15-base            libalgorithm-diff-xs-perl  libcrypt-dev  libgcc-15-dev           libitm1     libstdc++-15-dev  make               python3.14-venv
  cpp-15           fakeroot                 g++-x86-64-linux-gnu     gcc-15-x86-64-linux-gnu  libalgorithm-merge-perl  libdpkg-perl  libgomp1                liblsan0    libtsan2         python3-pip-whl

Suggested packages:
  bzip2-doc  gcc-15-locales  debian-keyring           g++-multilib     gcc-15-doc    autoconf  libtool  bison  gcc-doc         gdb-x86-64-linux-gnu  libstdc++-15-doc
  cpp-doc    cpp-15-doc      debian-tag2upload-keyring  g++-15-multilib  gcc-multilib  automake  flex     gdb    gcc-15-multilib  bzr                   make-doc

Recommended packages:
  python3-dev

Summary:
  Upgrading: 0, Installing: 45, Removing: 0, Not Upgrading: 50
  Download size: 76.7 MB
  Space needed: 260 MB / 5,747 MB available

Get:1 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15-base amd64 15.2.0-16ubuntu1 [39.0 kB]
Get:2 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libisl23 amd64 0.27-1build1 [691 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libmpc3 amd64 1.3.1-3 [54.6 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 cpp-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [12.9 MB]
Get:5 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 cpp-15 amd64 15.2.0-16ubuntu1 [1,030 B]
Get:6 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 cpp-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [5,840 B]
Get:7 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 cpp amd64 4:15.2.0-5ubuntu1 [22.4 kB]
Get:8 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libcc1-0 amd64 16-20260322-1ubuntu1 [51.2 kB]
Get:9 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libgomp1 amd64 16-20260322-1ubuntu1 [162 kB]
Get:10 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libitm1 amd64 16-20260322-1ubuntu1 [30.0 kB]
Get:11 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libasan8 amd64 16-20260322-1ubuntu1 [3,184 kB]
Selecting previously unselected package gcc-15-base:amd64.mdd64 liblsan0 amd64 16-20260322-1ubuntu1 [1,394 kB]
(Reading database… 139925 files and directories currently installed.)
Preparing to unpack …/00-gcc-15-base_15.2.0-16ubuntu1_amd64.deb…
Unpacking gcc-15-base:amd64 (15.2.0-16ubuntu1)…
Selecting previously unselected package libisl23:amd64.
Preparing to unpack …/01-libisl23_0.27-1build1_amd64.deb…
Unpacking libisl23:amd64 (0.27-1build1)…
Selecting previously unselected package libmpc3:amd64.
Preparing to unpack …/02-libmpc3_1.3.1-3_amd64.deb…
Unpacking libmpc3:amd64 (1.3.1-3)…
Selecting previously unselected package cpp-15-x86-64-linux-gnu.
Preparing to unpack …/03-cpp-15-x86-64-linux-gnu_15.2.0-16ubuntu1_amd64.deb…
Unpacking cpp-15-x86-64-linux-gnu (15.2.0-16ubuntu1)…
Selecting previously unselected package cpp-15.
Preparing to unpack …/04-cpp-15_15.2.0-16ubuntu1_amd64.deb…
Unpacking cpp-15 (15.2.0-16ubuntu1)…
Selecting previously unselected package cpp-x86-64-linux-gnu.
Preparing to unpack …/05-cpp-x86-64-linux-gnu_4%3a15.2.0-5ubuntu1_amd64.deb…
Unpacking cpp-x86-64-linux-gnu (4:15.2.0-5ubuntu1)…
Selecting previously unselected package cpp.
Preparing to unpack …/06-cpp_4%3a15.2.0-5ubuntu1_amd64.deb…
Unpacking cpp (4:15.2.0-5ubuntu1)…
Selecting previously unselected package libcc1-0:amd64.
Preparing to unpack …/07-libcc1-0_16-20260322-1ubuntu1_amd64.deb…
```

```
hina-ali32@ubuntu:~$ cd ~/Databases
hina-ali32@ubuntu:~/Databases$ python3 -m venv .venv
hina-ali32@ubuntu:~/Databases$ source .venv/bin/activate
(.venv) hina-ali32@ubuntu:~/Databases$ python -m pip install -r requirements.txt
Collecting pymongo==4.19.0 (from -r requirements.txt (line 2))
  Downloading pymongo-4.19.0-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (10 kB)
Collecting mysql-connector-python==9.7.0 (from -r requirements.txt (line 3))
  Downloading mysql_connector_python-9.7.0-cp314-cp314-manylinux_2_28_x86_64.whl.metadata (11 kB)
Collecting dnspython==2.9.0 (from -r requirements.txt (line 4))
  Downloading dnspython-2.9.0-py3-none-any.whl.metadata (5.7 kB)
Downloading pymongo-4.19.0-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (1.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.1/1.1 MB 519.2 kB/s eta 0:00:00
Downloading dnspython-2.9.0-py3-none-any.whl (354 kB)
Downloading mysql_connector_python-9.7.0-cp314-cp314-manylinux_2_28_x86_64.whl (21.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 21.7/21.7 MB 389.6 kB/s eta 0:00:00
Installing collected packages: mysql-connector-python, dnspython, pymongo
Successfully installed dnspython-2.9.0 mysql-connector-python-9.7.0 pymongo-4.19.0
(.venv) hina-ali32@ubuntu:~/Databases$
```

Task 3: Start the Database Services

```
(.venv) hina-ali32@ubuntu:~/Databases$ docker compose ps
NAME                    IMAGE       COMMAND                  SERVICE      CREATED              STATUS                        PORTS
databases-mongo-dev-1   mongo:8.0   "docker-entrypoint.s…"   mongo-dev    2 minutes ago        Up 2 minutes (healthy)        127.0.0.1:27017->27017/tcp
databases-mongo-prod-1  mongo:8.0   "docker-entrypoint.s…"   mongo-prod   About a minute ago   Up About a minute (healthy)   127.0.0.1:27019->27017/tcp
databases-mongo-test-1  mongo:8.0   "docker-entrypoint.s…"   mongo-test   About a minute ago   Up About a minute (healthy)   127.0.0.1:27018->27017/tcp
databases-mysql-1       mysql:8.4   "docker-entrypoint.s…"   mysql        2 minutes ago        Up 2 minutes (healthy)        33060/tcp, 127.0.0.1:3307->3306/tcp
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the MySQL and three MongoDB services running and reported as healthy.

```
(.venv) hina-ali32@ubuntu:~/Databases$ docker compose port mysql 3306
127.0.0.1:3307
(.venv) hina-ali32@ubuntu:~/Databases$ docker compose port mongo-dev 27017
127.0.0.1:27017
(.venv) hina-ali32@ubuntu:~/Databases$ docker compose port mongo-test 27017
127.0.0.1:27018
(.venv) hina-ali32@ubuntu:~/Databases$ docker compose port mongo-prod 27017
127.0.0.1:27019
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the host ports mapped to the four database containers.

Task 4: MySQL with Values in the Code

```
(.venv) hina-ali32@ubuntu:~/Databases$ nano db_connect.py
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MySQL: school_mysql
Server: 127.0.0.1 3307
Environment: dev
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows a successful MySQL connection to school_mysql on port 3307 using values written directly in db_connect.py, with the dev environment returned.

Task 5: MongoDB with Values in the Code

```
(.venv) hina-ali32@ubuntu:~/Databases$ nano db_connect.py
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows a successful MongoDB connection to school_dev on port 27017 using values written directly in the code.

Task 6: MongoDB with os.getenv

```
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the connection to school_dev on port 27017 using values read from environment variables.

```
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test

(.venv) hina-ali32@ubuntu:~/Databases$
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the same script connecting to school_test on port 27018 after only the exported values changed.

```
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the same script connecting to school_prod on port 27019 after only the exported values changed.

Task 7: MongoDB with a Credentials File

```
(.venv) hina-ali32@ubuntu:~/Databases$ cp -n credentials.ini.example credentials.ini
cp: warning: behavior of -n is non-portable and may change in future; use --update=none instead
(.venv) hina-ali32@ubuntu:~/Databases$ ls credentials.ini
credentials.ini
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows credentials.ini created from the supplied example file.

```
(.venv) hina-ali32@ubuntu:~/Databases$ chmod 600 credentials.ini
(.venv) hina-ali32@ubuntu:~/Databases$ stat -c '%a %n' credentials.ini
600 credentials.ini
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows credentials.ini restricted to mode 600, readable and writable only by its owner.

```
(.venv) hina-ali32@ubuntu:~/Databases$ export APP_ENV=dev
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) hina-ali32@ubuntu:~/Databases$
```

\

Shows the connection to school_dev on port 27017 using settings read from the credentials file with APP_ENV=dev.

```
(.venv) hina-ali32@ubuntu:~/Databases$ export APP_ENV=test
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the connection to school_test on port 27018 after only APP_ENV changed to test.

```
(.venv) hina-ali32@ubuntu:~/Databases$ export APP_ENV=prod
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod

(.venv) hina-ali32@ubuntu:~/Databases$
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the connection to school_prod on port 27019 after only APP_ENV changed to prod.

Exam Evaluation Questions

1. A Missing Environment Variable

```
(.venv) hina-ali32@ubuntu:~/Databases$ unset MONGO_DATABASE
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Set all four MongoDB environment variables first.
(.venv) hina-ali32@ubuntu:~/Databases$
```

Shows the program stopping with the missing-variable message after MONGO_DATABASE was removed.

Prediction: The program would stop with Set all four MongoDB environment variables first. and make no connection.

Result: The message was printed after MONGO_DATABASE was removed. After MONGO_DATABASE=school_prod was restored, the program connected to school_prod on port 27019.

Explanation: os.getenv("MONGO_DATABASE") returned None because the variable no longer existed. database_name became empty, so the if condition was true and SystemExit ended the program before the URI was built.

2. Which Source Does the File Version Read?

```
(.venv) hina-ali32@ubuntu:~/Databases$ unset APP_ENV
(.venv) hina-ali32@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) hina-ali32@ubuntu:~/Databases$
```

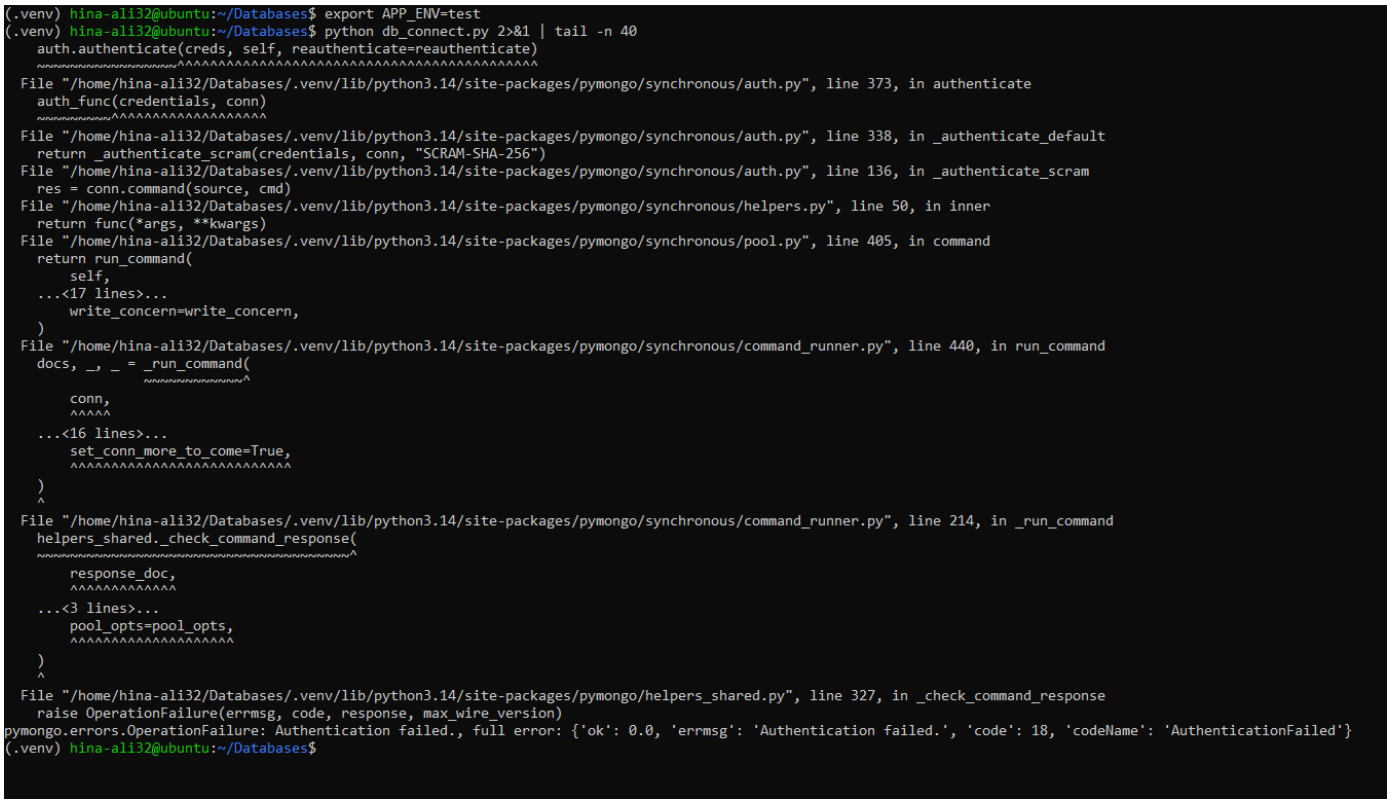
Shows the dev connection made even though the prod variables were still exported, because the file version reads only credentials.ini and APP_ENV.

Prediction: The program would connect to dev, not prod.

Result: With APP_ENV unset, the program connected to school_dev on port 27017. With APP_ENV=test, it connected to school_test on port 27018.

Explanation: The file version ignores the exported MONGO_* variables and reads only APP_ENV. With APP_ENV unset, os.getenv("APP_ENV", "dev") returns dev, so the [dev] section of credentials.ini supplied the settings.

3. A Wrong Password in the File



Shows the authentication error produced by a wrong password in the test section of the credentials file.

(output shortened with tail because the full traceback is long).

Prediction: The client would reach the test endpoint, but authentication would fail.

Result: With password = wrong, the program stopped with an authentication error. After the dummy password was restored, it connected to school_test again, and APP_ENV was set back to prod.

Explanation: The endpoint only decides which server is contacted. MongoDB then checks the username and password against the stored user, so a wrong password fails even when the endpoint is correct.